

Blood Donation System

Full-Stack Project Details & Workflow Overview

Project Stack: HTML, CSS, JavaScript, PHP, MySQL

Goal: A complete role-based system (Admin & User) with secure conditions and 90-day donation rules.

Phase 1: Public Interface (Frontend)

Ei page-gulo jekono normal visitor dekhte parbe.

- **Home Page (`index.php`):**

- Ekta sundor Hero Section jekhane "Donate Blood, Save Life" theme thakbe.
- **Live Stats:** Kotojon donor registered, kotogulo successful donation (DB theke count).
- **Recent Requests:** Urgent blood request gular slider/list (shudhu Admin approved).

- **Search Donors Page:**

- Visitor ra blood group aur location/district diye search korte parbe.
- *Condition:* Donor der list show korbe, kintu phone number/email hide kora thakbe. Visitor ke bolbe "Login to view contact details".

- **About Us & Contact Us:** Project details, team info, aur contact form (submit korle admin er kache jabe).

Phase 2: Authentication System

Ekhane PHP aur MySQL er main interaction shuru.

- **User Registration (`register.php`):**

- *Inputs:* Name, Email, Password, Blood Group, Phone, District, Age, **Last Donation Date**.
- *Condition:* JS diye check korbe password same ki na. PHP check korbe email unique ki na. Age 18+ hote hobe.

- **Login Page (`login.php`):**

- Email aur Password diye login.
- *Condition (Crucial):* PHP check korbe `role` ki. `role == 'admin'` hole admin dashboard, r `role == 'user'` hole user dashboard e nibe.

Phase 3: User Panel (Limited Access)

Login korar por ekjon sadharon user ei feature-gulo access korbe.

- **User Dashboard Home:** Simple overview - total past donations, active blood requests.
- **Profile & Settings:**
 - User tar details update korte parbe.
 - *Main Feature:* User tar "Last Donation Date" update korte parbe jodi se baire kothao blood diye thake.
- **Post a Blood Request (`request_blood.php`):**
 - *Inputs:* Patient Name, Hospital Address, Required Blood Group, Units (bags), Date needed, Urgency.
 - *Workflow:* Submit korle directly public hobe na. Database-e status `Pending` thakbe. Admin approve korle public hobe.
- **My Requests History:** User nijer kora shob request-er status (Pending/Approved/Fulfilled) check korte parbe.
- **Available Blood Requests (Feed):**
 - Onno user-der kora shob *Approved* request thakbe.
 - Protita request-er pashe **"I Want to Donate"** button thakbe.
 - *Condition (Self-block):* Nijer kora request-e "Donate" button e click korle error dibe.
 - *Condition (90 Days Rule):* Click korle PHP check korbe ajker date theke tar last donation date er gap 90 days hoyeche ki na. Na hole system error dibe: "You are not eligible to donate yet."
- **My Donation History:** Kobe kobe kake blood diyeche tar list.

Phase 4: Admin Panel (Full Access)

Admin dashboard-e total control thakbe.

- **Admin Dashboard Home:** Widgets - Total Users, Total Pending Requests, Total Successful Donations.
- **Manage Users:** Database-er shob user er list. Admin chaile fake account delete ba spammer ke ban korte parbe.
- **Manage Blood Requests (Moderation):**
 - Ekhane shob `Pending` request asbe.
 - Admin request verify kore **Approve** ba **Reject** korbe. Approve korlei eta User Feed-e jabe.

- **Manage Donations (Track Progress):**

- Keu "I Want to Donate" click korle ekhane notification asbe.
- Blood dewa complete hoye gele, Admin status change kore **Completed** kore dibe. Tokhon automatically oi user-er "Last Donation Date" ajker date hishebe update hoye jabe.

Phase 5: Technical & Security Features

Assignment-e perfect kaj korar jonno ei bishoygulo thakte hobe:

- **Data Integrity:** Ekhane list kora 1 line code ba logic o bad dewa jabe na. Full architecture perfectly code-e maintain korte hobe.
- **Session Security:** Normal user URL bar-e `admin_dashboard.php` likhle access hobe na, login page e kick out korbe.
- **SQL Injection Prevention:** Database query-te `mysqli_real_escape_string` ba PDO prepared statements use korte hobe.
- **UI/UX Design:** CSS (Tailwind/Bootstrap) use kore site ke gorgeous o professional kora. Eye-catching headers o clean container layouts babohar korte hobe jeno user experience bhalo hoy.